Chapter 5

Inverses and Radicals

A function sends inputs forward to outputs. This week we ask when that motion can be reversed. If an output remembers exactly which input produced it, then the function can be undone. If two different inputs produce the same output, the reverse direction has a choice to make, and that choice prevents the reverse relation from being a function.

This idea connects several topics that may at first look separate. Domain and range tell us where a function starts and where it lands. One-to-one functions are the functions whose outputs do not get shared. Inverses run one-to-one functions backward. Radicals, such as square roots and cube roots, are inverses of power functions, so their domains come from the same reversing process. The goal is to see these as one story rather than as four unrelated rules.

5.1 Domain and Range

Before a function can be reversed, we need to know its input set and output set. The *domain* is the set of inputs the function is allowed to accept. The *range* is the set of outputs the function actually produces. These two sets are part of the function's meaning. A formula by itself tells us how to compute, but the domain and range tell us where the computation is allowed to live.

For a function given by a formula, the natural domain is every real input for which the formula is defined. The range is usually read from the graph, from transformations, or from the nature of the formula. In this course the most common domain restrictions come from two operations: division and even roots.

Definition: Domain and range. The *domain* of f is the set of all inputs x for which $f(x)$ is defined. The *range* is the set of all outputs $f(x)$ produced as x ranges over the domain.

Division forbids a zero denominator, because division by zero is not defined. An even root forbids a negative radicand, because the even root of a negative number is not real. Everything else we use here, such as polynomials, odd roots, and absolute values, accepts all real inputs unless an outside restriction is stated.

Key idea. To find a natural domain from a formula, look for forbidden operations:

- denominators cannot equal zero;
- radicands under even roots must be nonnegative;
- odd roots, polynomials, and absolute values do not create domain restrictions by themselves.

The range then records the outputs that actually occur after the domain is chosen.

Example 5.1.1. *(reading the domain from the formula)* For $f(x) = \dfrac{1}{x-5}$ the denominator vanishes at $x = 5$, so the domain is all reals except 5. For $g(x) = \sqrt{x-3}$ the radicand $x - 3$ must be nonnegative, so the domain is $x \geq 3$. For $h(x) = x^2 + 1$ no operation restricts the input, so the domain is all reals, while the range is $y \geq 1$ because a square is never negative.

The important point in the example is that the domain is not guessed from the general shape. It is forced by the operations in the formula. The range asks a different question: after the allowed inputs are used, which outputs appear?

Common pitfall. Domain restrictions come from the operations, not from the graph's appearance. An odd root such as $\sqrt[3]{x}$ accepts negatives, since a cube root of a negative number is a real negative number; only *even* roots restrict the radicand.

5.2 One-to-One and Onto

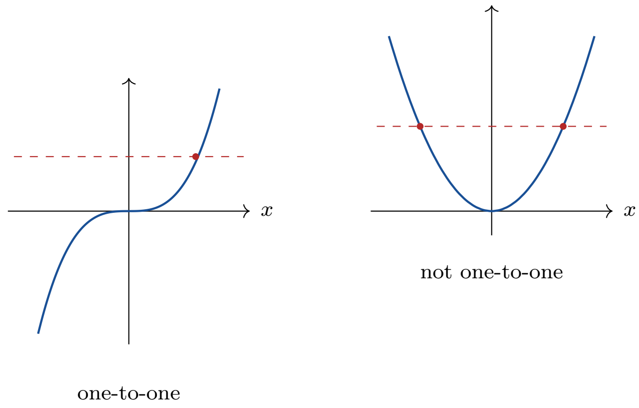
Now we turn from the question "where is the function defined?" to the question "can its arrows be reversed?" A function already has the rule that each input goes to exactly one output. For an inverse function, we need the reverse assignment also to be single valued: each output must point back to exactly one input.

Two properties describe how a function distributes its inputs over its outputs. The first, one-to-one, says that no two inputs share an output. The second, onto, says that every output in a chosen target set is hit by the function.

Definition: One-to-one and onto. A function f is *one-to-one* if distinct inputs always give distinct outputs: $x_1 \neq x_2$ forces $f(x_1) \neq f(x_2)$. It is *onto* a target set of outputs if every value in that set is attained by some input.

One-to-one is the property that matters most for inverses in this course. If an output is shared, the reverse direction cannot decide which input to choose. The graph gives a clean test, because a horizontal line collects all points with the same output value.

> **Key idea. Horizontal line test.** A function is one-to-one if and only if no horizontal line crosses its graph more than once. A repeated crossing marks two inputs sharing one output, which violates the one-to-one property.



Example 5.2.1. *(applying the horizontal line test)* The line $f(x) = 2x - 1$ is one-to-one; every horizontal line meets it once. The parabola $f(x) = x^2$ is not one-to-one; the horizontal line $y = 4$ meets it at $x = 2$ and $x = -2$, two inputs sharing the output 4. Restricting the parabola to $x \geq 0$ removes the left half and makes the restricted function one-to-one.

The restriction in the last sentence is not a trick. It is how many inverses are created: if the original function repeats outputs, we sometimes choose a domain where the repeats are removed. The square root function comes from exactly this choice.

Whether a function is onto depends on the target set declared for its outputs. The squaring function is onto the nonnegative reals (every nonnegative number is a square) but not onto all reals (no input squares to a negative number). Choosing the target set to be exactly the range makes any function onto by construction.

5.3 The Inverse Function

When f is one-to-one, each output came from exactly one input, so the assignment can be reversed: send each output back to the input that produced it. That reverse assignment is the inverse function f^{-1}. The notation looks like an exponent, but here it means "inverse function," not reciprocal.

Definition: Inverse function. If f is one-to-one, its *inverse* f^{-1} is the function satisfying

$$f^{-1}(y) = x \quad \text{exactly when} \quad f(x) = y.$$

Equivalently, $f^{-1}(f(x)) = x$ for every x in the domain of f, and $f(f^{-1}(y)) = y$ for every y in the range.

Key idea. An inverse undoes the function. Its domain is the range of f and its range is the domain of f; the two sets trade places, because reversing the arrows swaps inputs and outputs. Only a one-to-one function has an inverse, since a shared output would have no single input to return to.

5.3.1 Finding an inverse

Because the inverse swaps the roles of input and output, the algebra is: write $y = f(x)$, exchange x and y, and solve for y. This procedure is not just a formal recipe. The swap represents the reversal of the arrows. Solving for y puts the reversed rule back into function notation.

Example 5.3.1. *(finding an inverse)* Invert $f(x) = 2x - 1$. Write $y = 2x - 1$, swap to $x = 2y - 1$, and solve:

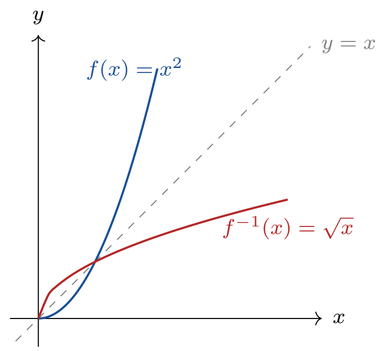
$$y = \frac{x+1}{2}.$$

So $f^{-1}(x) = \dfrac{x+1}{2}$. Check:

$$f^{-1}(f(x)) = \frac{(2x-1)+1}{2} = x.$$

The check confirms that applying f and then f^{-1} brings the input back to where it started.

The picture. Swapping x and y reflects the graph across the line $y = x$. The graph of f^{-1} is the mirror image of the graph of f in that diagonal, because every point (a, b) on f becomes the point (b, a) on f^{-1}.



The picture also shows why domain restrictions matter. The full parabola $y = x^2$ would reflect to a sideways parabola, which fails the vertical line test. The restricted right half reflects to the square-root graph, which is a function.

5.4 Radicals as Inverses of Powers

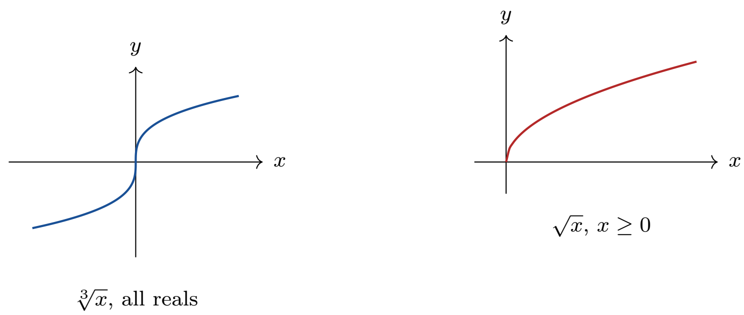
A radical is the inverse of a power function. The square root inverts squaring, the cube root inverts cubing, and in general $\sqrt[n]{x}$ inverts x^n. The parity of n decides whether the power function repeats outputs. Odd powers keep moving in one direction, so they are one-to-one on all real numbers. Even powers fold the line across the y-axis, sending x and $-x$ to the same output, so they must be restricted before being inverted.

Key idea. Radicals and parity.

- *n odd.* x^n is one-to-one on all reals, so $\sqrt[n]{x}$ is defined for all reals; negatives are allowed.
- *n even.* x^n is not one-to-one on all reals (it shares outputs across $\pm x$), so it must be restricted to $x \geq 0$ before inverting. Its inverse $\sqrt[n]{x}$ then has domain and range $x \geq 0$.

Example 5.4.1. *(why the square root needs a restriction)* The function x^2 fails the horizontal line test on all reals. Restricted to $x \geq 0$ it is one-to-one, and its inverse is $\sqrt{x}$, with domain $x \geq 0$ and range $y \geq 0$. The discarded left half $x \leq 0$ would have inverted to $-\sqrt{x}$; the two halves are the two branches of the sideways parabola, only one of which is a function.

Example 5.4.2. *(an odd root accepts negatives)* The function x^3 is one-to-one on all reals, so $\sqrt[3]{x}$ is defined everywhere. For instance $\sqrt[3]{-8} = -2$, since $(-2)^3 = -8$. No restriction is needed because cubing never repeats an output.



5.4.1 Solving equations with radicals

A radical equation is solved by isolating the radical and raising both sides to the matching power. The solving step is allowed, but it is not always reversible. Raising an even root to an even power can introduce a false solution, because the squaring step forgets whether the original right-hand side was nonnegative. That is why every candidate must be checked in the original equation.

Example 5.4.3. *(solving and checking)* Solve $\sqrt{x+5} = x - 1$. Square both sides:

$$x + 5 = x^2 - 2x + 1,$$

so $x^2 - 3x - 4 = 0$, giving $(x-4)(x+1) = 0$ and candidates $x = 4$ and $x = -1$. Check in the original: $x = 4$ gives $\sqrt{9} = 3$ and $4 - 1 = 3$, valid. $x = -1$ gives $\sqrt{4} = 2$ but $-1 - 1 = -2$; the two sides disagree, so $x = -1$ is rejected. The only solution is $x = 4$.

Common pitfall. Squaring both sides can manufacture solutions the original equation never had, because $\sqrt{\ }$ returns the nonnegative root while squaring forgets the sign. Always substitute each candidate back into the original radical equation and discard any that fail.

5.5 Exercises

Exercises 5.5

5.5.1. State the domain of each: (a) $\dfrac{x}{x^2 - 9}$; (b) $\sqrt{2x-6}$; (c) $\sqrt[3]{x+1}$; (d) $\dfrac{1}{\sqrt{x-4}}$.

5.5.2. Give the range of each, with a reason: (a) $f(x) = x^2 - 4$; (b) $f(x) = \sqrt{x} + 2$; (c) $f(x) = -|x|$.

5.5.3. Apply the horizontal line test to decide which are one-to-one: (a) $f(x) = 3x + 2$; (b) $f(x) = x^2 - 1$; (c) $f(x) = x^3$; (d) $f(x) = |x|$.

5.5.4. Explain, using the definition, why a function must be one-to-one to have an inverse. What goes wrong at a shared output?

5.5.5. Find the inverse of each and state its domain: (a) $f(x) = 3x - 7$; (b) $f(x) = \dfrac{x+2}{5}$; (c) $f(x) = x^3 + 1$.

5.5.6. For $f(x) = x^2$ restricted to $x \geq 0$, find f^{-1}, verify $f^{-1}(f(x)) = x$ on the restricted domain, and state the domain and range of f^{-1}.

5.5.7. Explain why $\sqrt[3]{x}$ is defined for all real numbers while $\sqrt{x}$ is not, in terms of the parity of the power being inverted.

5.5.8. Sketch $f(x) = x^2$ on $x \geq 0$ and its inverse on the same axes, and show that each is the reflection of the other across $y = x$.

5.5.9. Solve and check, discarding any false solutions: (a) $\sqrt{2x+3} = x$; (b) $\sqrt{x+7} = x + 1$.

5.5.10. For an increasing one-to-one function, intersections of the graph of f with the graph of f^{-1} occur on the line $y = x$. Explain this using the reflection picture, and contrast it with a decreasing self-inverse function such as $f(x) = 1 - x$.

5.5.11. The function $f(x) = \sqrt{x-2} + 1$ is built from the parent $\sqrt{x}$ by transformations. State the domain and range, find the inverse, and confirm the domain of the inverse equals the range of f.

5.5.12. Determine whether $f(x) = \dfrac{1}{x}$ is its own inverse by computing f^{-1}, and explain the symmetry of its graph in light of the result.